



■ CS177 Bioinformatics: Software Development and Applications

Lecture 13: Multiple Sequence Alignment (MSA)

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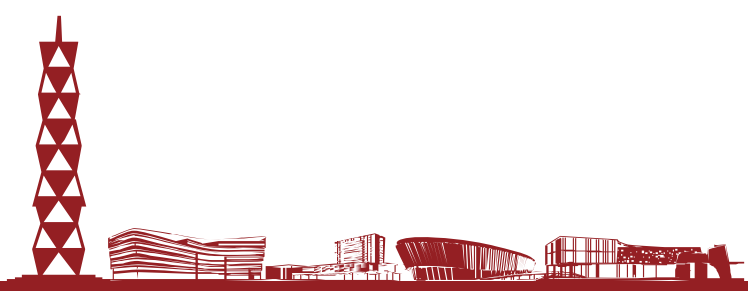
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Learning objectives



- At the end of this lecture, you will be able to:
 - Give the definition and describe the features of multiple sequence alignment (MSA)
 - Explain two main scoring functions for evaluating MSA (sum-of-pairs, distance to consensus)
 - List a few exact and heuristic MSA algorithms
 - Describe the algorithm of ClustalW
 - Describe an example of deep learning on MSA data for protein structure prediction: MSA Transformer





Introduction





Multiple sequence alignment: definition



- Multiple Sequence Alignment (MSA)
 - Generalized pairwise sequence alignment to include more than 2 sequences of protein (or nucleic acid)
- **Input:**
 - A set of more than 2 sequences
 - A method for scoring an alignment
- **Output:**
 - Determined correspondences between the sequences such that **the similarity is maximized**
- Note:
 - Sequences are partially or completely aligned
 - Homologous residues (or nucleotides) are aligned in columns across the length of the sequences
 - Global Alignment



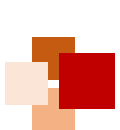


Example: 5 alignments of 5 globins



- Let's look at the multiple sequence alignment (MSA) of five globins proteins. We use 5 prominent MSA programs (where each program offers unique strengths):
 - ClustalW
 - Praline
 - MUSCLE (used at HomoloGene)
 - ProbCons
 - T-COFFEE
- Let us focus on a histidine (H) residue that has a critical role in binding oxygen in globins and **should** be aligned. But often it's not aligned, and all the five programs give different answers
- What conclusion would you make from this example?





ClustalW

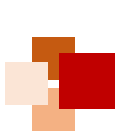


CLUSTAL W (1.83) multiple sequence alignment

```
beta globin  -----MVHLTPEEKSAVTALWGKVNV--EVGGEALGRLLVVYPWTQRFFESFG- 47
myoglobin    -----MGLSDGEWQLVLNVWGKVEADIPGHGQEV LIRLFKGH PETLEKFDKFK- 48
neuroglobin  -----MERPEPELI RQSWRAVSRSPLEHGT VLFARLFALEPDLLPLFQYNCR 47
soybean      -----MVAFT EKQDALVSSSF EAFKANIPQYSVV FYTSILEKAPAAKD LFSFLA- 49
rice         MALVEDNNAVAVSFS EEQEALVLKSWAILKKDSANIALRFFLKI FEVAPSASQMFSFLR- 59
               :   :   :   :   . . .   .   : :   *   * .
               ▼
beta globin  DLSTPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLS-----ELHCDKLHVDPE 102
myoglobin    HLKSEDEMKA SEDLK KHGATVLTALGGILKKKGHHEAEIKPLA-----QSHATKHKIPVK 103
neuroglobin  QFSSPEDCLSSPEFLDHIRKVMLVIDAAVTNVEDLSSLEEYLA S---LGRKHRAVG VKLS 104
soybean      --NGVDPT--NPKLTGHA EKL FALVRDSAGQLKASGTVVADAA----LGSVHAQKAVTDP 101
rice         --NSDVPLEKNPKLKT HAMS VFVMTCEAA AQLRKAGKVTVRD TTLKRLGATHLKYGVGDA 117
               .   . . . *   . : :   :   :   :   :
               :   :   :   :   :   :   :   :   :   :
beta globin  NFRLLGNVLVCVLAH HF-GKEFTPPVQAAYQKV VAGVANALAHKYH----- 147
myoglobin    YLEFISECIIQVLQSKH-PGDFGADAQGAMNKALELFRKDMASNYKELGFQG 154
neuroglobin  SFSTVGESLLYMLEKCL-GPAFTPATRAAWSQLYGAVVQAMSRGWDGE---- 151
soybean      QFVVVKEALLKTIKAAV-GDKWSELSRAWEVAYDELA AAIKKA----- 144
rice         HFEVVKFALLDTIKEEVPADMWSPAMKSAWSEAYDHLVAAIKQEMKPAE--- 166
               :   :   : :   :   :   :   *   .   .   :
```

Note how the region of a conserved histidine (▼) varies depending on which of the five prominent algorithms is used.





(a) Praline multiple sequence alignment

```
beta globin      .....MVHLTPEEKSAVTALWGKV..NVDEVGGEALGRLLVVYPWTQRFFES.FG
myoglobin       .....MGLSDGEWQLVLNVWGKVEADIPGHGQEV LIRLFKGHPETLEKFDK.FK
neuroglobin     .....MERPEPELIRQSWRAVSRSPLEHGT VLFARLFALEPDLLPLFQYNCR
soybean         .....MVAFTEKQDALVSSSF EAFKANIPQYSVVFYTSILEKAPAAKDLFS..FL
rice            MALVEDNNAVAVSFSEEQEALVLKSWAILKSDANIALRFFLKI FEVAPSASQMFS..FL
Consistency     000000000014265438257934573463364343624453686433*35344*50063
```

```
beta globin      DLSTPDAVMGNPKVKAHGKKVLGAFSDG LAHLDNLKGT FATLSEL..HCDKLH....VDP
myoglobin       HLKSEDEMKA SEDLKKHGATVLTALGGILKKKGHHEAEIKPLAQS..HATKHK....IPV
neuroglobin     QFSSPEDCLSSPEFLDHIRKVMLVIDAAVTNVEDLSSLEEYLA SLGRKHRAVG....VKL
soybean         A.NGVDP..TNPKLTGHA EKLFALVRDSAGQL.KASGTVVADAA...LGSVHAQKAVTD
rice            R.NSDVPLEKNPKLKTHAMSVFVMTCEAAAQL.RKAGKVTVRDTTLKRLGATHLKYGVGD
Consistency     3166354224776653*4368635424454451335634333542003335440000922
```

```
beta globin      ENFRLLGNVLVCVLAHHF.GKEFTPPVQAAYQKV VAGVANALAHKYH.....
myoglobin       KYLEFISECIIQVLQSKH.PGDFGADAQGAMNKALELFRKDMASNYKELGFQG
neuroglobin     SSFSTVGESLLYMLEKCL.GPAFTPATRAAWSQLYGAVVQAMSRGWD..GE..
soybean         PQFVVVKEALLKTIKAAV.GDKWSD ELSRAWEVAYDELA AAIKKA.....
rice            AHFEVVKFALLDTIKEEVPADMWSPAMKSAWSEAYDHLVAAIKQEMKPAE...
Consistency     43744844498258542305336554454*55465426446754322001000
```

Note also the changing pattern of gaps within the boxed region in these five different alignments.





MUSCLE

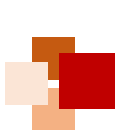


(b) MUSCLE (3.6) multiple sequence alignment

```

beta globin  -----MVHLTPEEKSAVTALWGKVNVD--EVGGEALGRLLVVYPWTQRFFES-FG
myoglobin   -----MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIRLFKGHPETLEKFDK-FK
neuroglobin -----MERPEPELIRQSWRAVSRSPLEHGTVLFARLFALEPDLLPLFQYNCR
soybean      -----MVAFTEKQDALVSSSFEAFKANIPQYSVVFYTSILEKAPAAKDLFSF-LA
rice         MALVEDNNAVAVSFSEEQEALVLKSWAILKKDSANIALRFFLKIFEVAPSASQMFSF-LR
               :   :   :   :   .   .   .   :   *   *
               ▽
beta globin  DLSTPDAVMGNPKVKAHGKKVLGAF---SDGLAHLDNLKGTFATLSELHCDKLH--VDPE
myoglobin   HLKSEDEMKASEDLKKHGATVLTAL---GGILKKKGHHEAEIKPLAQSHATKHK--IPVK
neuroglobin QFSSPEDCLSSPEFLDHIRKVMLVI---DAAVTNVEDLSSLEEYLASLGRKHRAVGVKLS
soybean      NGVDP----TNPKLTGHAEKLFALVRDSAGQLKASGTVVAD----AALGSVHAQKAVTDP
rice         NSDVP--LEKNPKLKTHAMSVFVMTCEAAQLRKAGKVTVRDTTLKRLGATHLKYGVGDA
               . . . * . : :           :           :
               ▽
beta globin  NFRLLGNVLVCVLAHHFGKE-FTPPVQAAYQKVVAGVANALAHKYH-----
myoglobin   YLEFISECIIQVLQSKHPGD-FGADAQGAMNKALELFRKDMASNYKELGFQG
neuroglobin SFSTVGESLLYMLEKCLGPA-FTPATRAAWSQLYGAVVQAMSRGWDGE----
soybean      QFVVVKEALLKTIKAAVGDK-WSELSRAWEVAYDELAAAIKKA-----
rice         HFEVVKFALLDTIKEEVPADMWSPAMKSAWSEAYDHLVAAIKQEMKPAE---
               :   :   : :   :           *   .   .   :
    
```





Probcons



(c) PROBCONS

```
beta globin  M-----VHLTPEEKSAVTALWGKVNVDD--EVGGEALGRLLVVYPWTQRFFES-FG
myoglobin   M-----GLSDGEWQLVLNVWVGKVEADIPGHGQEVLIIRLFKGGHPETLEKFDK-FK
neuroglobin M-----ERPEPELIHQSWRAVSRSPLEHGTIVLFARLFALEPDLLPLFQYNCR
soybean      M-----VAFTEKQDALVSSSFEEAFKANIPQYSVVFYTSILEKAPAAKDLFSF-LA
rice         MALVEDNNAVAVSFS EEQEALVLKSWAILKKDSANIALRFFLKIFEVAPSASQMFSF-LR
*           * : : : : : . . . . . : : * *
```

```
beta globin  DLSTPDAVMGNPKVKAHGKKVLGAFSDGLAHLD---NLK---GTFATLSELHCDKLHVDP
myoglobin   HLKSEDEMKA SEDLKKHGATVLTALGGI---LKKKGHHE---AEIKPLAQSHATKHKIPV
neuroglobin QFSSPEDCLSSPEFLDHIRKVMLVIDAAVTNVEDLSSLE---EYLASLGRKHRAV-GVKL
soybean      NGVDP----TNPKLTGHA EKLFALVRDSAGQLKASGTVV---ADAALGSVHAQK-AVTD
rice         NSDVP--LEKNPKLKTHAMSVFVMTCEAA AQLRKAGKVTVRD TTKRLGATHLKY-GVGD
.           : . . . * . : : : : . . . * *
```

```
beta globin  ENFRLLGNVLVCVLAHHF-GKEFTPPVQAAYQKVVAGVANALAHK-----YH
myoglobin   KYLEFISECIIQVLQSKH-PGDFGADAQGAMNKALELFRKDMASNYKELGFQG
neuroglobin SSFSTVGESLLYMLEKCL-GPAFTPATRAAWSQLYGAVVQAMSRG---W-DGE
soybean      PQFVVVKEALLKTIKAAV-GDKWSELSRAWEVAYDELA AAIK-----KA
rice         AHFEVVKFALLDTIKEEVPADMWSPAMKSAWSEAYDHLVAAIKQE---MKPAE
:           : : : : : : * . . . :
```



(d)

CLUSTAL FORMAT for T-COFFEE Version_5.13

```
beta globin  -----MVHLTPEEKSAVTALWGKVNVDD--EVGGEALGRLLVVYPWTQRFFE-SFG
myoglobin   -----MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIIRLFKGGHPETLEKFD-KFK
neuroglobin -----MERPEPELIQSWRAVSRSPLEHGTVLFARLFALPDLLPLFQYNCR
soybean      -----MVAFTKQDALVSSSFQAFKANIPQYSVVFYTSILEKAPAAKDLFS-FLA
rice         MALVEDNNAVAVSFSSEQEALVLKSWAILKKDSANIALRFFLKIIFEVAPSASQMFS-FLR
               :   :   :   :   .   .   .   :   :   *   *   .
               ▽
beta globin  DLSTPDAVMGNPKVKAHGKKVLGAFSDGLAHLDNL---KGTF---ATLSELHCDKLHVDIP
myoglobin   HLKSEDEMKA SEDLKKHGATVLTAL---GGILKKKGHHEAE---IKPLAQSHATKHKIEV
neuroglobin QFSSPEDCLSSPEFLDHIRKVMLVIDAAVTNVEDL---SSLEEYLA SLGRKH-RAVGVKL
soybean      NGVDP----TNPKLTGHA EKLFALVRDSAGQLKASGTVVAD----AALGSVHAQKAVTDP
rice         NSDVP--LEKNPKLKTHAMSVFVMTCEAAAQLRKAGKVTVRDTTLKRLGATHLKYGVGDA
               .   .   .   *   .::               :   *   *
               :   :   :   :   :   :   :   :   :   :   :   :   :   :   :   :
beta globin  ENFRLLG NVLVCVLAH HF-GKEFTPPVQAAYQKV VAGVANALAHKYH-----
myoglobin   KYLEFISECIIQVLQSKH-PGDFGADAQGAMNKALELFRKDMASNYKELGFQG
neuroglobin SSFSTVGESLLYMLEKCL-GPAFTPATRAAWSQLYGAVVQAMSRGWDG----E
soybean      Q-FVVVKEALLKTIKAAV-GDKWSELSRAW EVAYDELAAAIKKA-----
rice         H-FEVVKFALLDTIKEEVPADMWSPAMKSAWSEAYDHLVAAIKQE---MKPAE
               :   :   :   :   :   :   :   :   :   :   :   :   :   :   :   :
```





Multiple sequence alignment: properties



- One conclusion from the example in the preceding slides is that **there is no single best approach to MSA**. Dozens of new programs have been introduced in recent years
- Not necessarily only one “correct” alignment of a protein family
- Protein sequences **evolve**, and the corresponding three-dimensional structures of proteins also **evolve**
- It may be impossible to identify amino acid residues that align properly (structurally) throughout an MSA
- For two proteins sharing 30% amino acid identity, about 50% of the individual amino acids are **superposable** in the two structures (i.e. they are structurally similar)





Multiple sequence alignment: features



- Some aligned residues, such as cysteines that form disulfide bridges, may be highly **conserved**
- There may be **conserved** motifs such as a transmembrane domain
- There may be **conserved** secondary structure features
- There may be regions with **consistent** patterns of **insertions or deletions** (i.e. **indels**)





Multiple sequence alignment: uses



- MSA is more sensitive than pairwise alignment to **detect homologs**
- BLAST output can take the form of an MSA, and can reveal conserved residues or motifs
- A single query can be searched against a database of MSAs (e.g. **PFAM**)
- Regulatory regions of genes may have **consensus sequences** identifiable by MSA





Scoring Function



Sum-of-Pairs function



- How do we assess the quality of an MSA?
- Usually, it is assumed that the alignment score is the sum of column scores. Therefore, we need a way to assign a score to each column and add them up to get the alignment score
- To score a column, we want a function with k arguments, where k is the number of sequences
- **Sum-of-Pairs function:** Sum of scores of pairwise alignments in each column added together



■ Sum-of-Pairs (SP) score



- $SP\text{-score}(\mathbb{I}, -, \mathbb{I}, \mathbb{V}) = p(\mathbb{I}, -) + p(\mathbb{I}, \mathbb{I}) + p(\mathbb{I}, \mathbb{V}) + p(-, \mathbb{I}) + p(-, \mathbb{V}) + p(\mathbb{I}, \mathbb{V})$
 - $p(a, b)$ is the pairwise score of symbols a and b , specified by a substitution matrix
 - $p(a, -)$ or $p(-, a)$ is a specified **gap penalty**
 - $p(-, -) = 0$
- SP-score is **independent** of the order of arguments, e.g.
$$SP\text{-score}(\mathbb{I}, -, \mathbb{I}, \mathbb{V}) = SP\text{-score}(\mathbb{V}, \mathbb{I}, \mathbb{I}, -)$$
- $k \cdot (k-1)/2$ pairwise scores need to be added for k arguments
- SP scoring system is widely used due to its simplicity and effectiveness





Sum-of-Pairs (SP)



- e.g. $s(x, y) = +1$ if $x = y$; and -1 if $x \neq y$; $g = -2$ (gap penalty)

- MSA M :

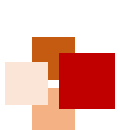
P	Y	R	F	T	-	-	-	I	K	S	M
P	Y	K	F	-	-	-	S	I	K	S	M
P	Y	M	Y	-	-	-	S	S	E	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

Column scores:

+6	0	-6	-4	-9	-6	-6	-3	-4	-4	+6	+6
----	---	----	----	----	----	----	----	----	----	----	----

$$SP\text{-score}(M) = -24$$





Sum-of-Pairs (SP)



- The SP-score of an MSA M of k sequences is the sum of the **induced pairwise alignments** (also called the *projection* of M):

$$SP - score(M) = \sum_{j=2}^k \sum_{i=1}^{j-1} score(M_{ij})$$

- where M_{ij} is the pairwise alignment induced by the sequences S_i and S_j



Sum-of-Pairs (SP)

- $s(x,y) = +1$ if $x = y$; and
- $s(x,y) = -1$ if $x \neq y$; $g = -2$
- **Induced pairwise alignments:**

$M_{1,2}$

P	Y	R	F	T	-	I	K	S	M
P	Y	K	F	-	S	I	K	S	M

$$\text{score}(M_{1,2}) = 2$$

$M_{1,3}$

P	Y	R	F	T	-	I	K	S	M
P	Y	M	Y	-	S	S	E	S	M

$$\text{score}(M_{1,3}) = -4$$

$M_{1,4}$

P	Y	R	F	T	-	-	-	I	K	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$$\text{score}(M_{1,4}) = -9$$

P	Y	R	F	T	-	-	-	I	K	S	M
P	Y	K	F	-	-	-	S	I	K	S	M
P	Y	M	Y	-	-	-	S	S	E	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$M_{2,3}$

P	Y	K	F	S	I	K	S	M
P	Y	M	Y	S	S	E	S	M

$$\text{score}(M_{2,3}) = 1$$

$M_{2,4}$

P	Y	K	F	-	-	-	S	I	K	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$$\text{score}(M_{2,4}) = -7$$

$M_{3,4}$

P	Y	M	Y	-	-	-	S	S	E	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$$\text{score}(M_{3,4}) = -7$$

$$SP\text{-score}(M) = 2 - 4 - 9 + 1 - 7 - 7 = -24$$

Minimizing distance to consensus

- Another measure for scoring an MSA is the **distance to consensus**
- Let $S = (s_1', s_2', \dots, s_k')$ be a multiple alignment of strings $s_1, s_2, \dots, s_k$, a string $c = c_1 \dots c_l$ is called a **consensus** of S , if c_j (for $j = 1, \dots, l$) is the symbol of the highest frequency in the j th column in S (i.e. **majority voting**)
- The **distance of an alignment to a consensus** is defined as the sum of numbers of characters in S that are different from the consensus character in each column
- A smaller distance to consensus means a better alignment
- Example: Consider the following alignment of the strings $s_1 = \text{AATGCT}$, $s_2 = \text{ATTC}$ and $s_3 = \text{TCC}$:

s_1'	=	A	A	T	G	C	T
s_2'	=	A	-	T	T	C	-
s_3'	=	-	-	-	T	C	C
c	=	A	A	T	T	C	T

- Then, c is a consensus of this alignment, and $\text{distance}(c, (s_1', s_2', s_3')) = 1 + 2 + 1 + 1 + 0 + 2 = 7$



Exact and Heuristic Methods



Exact methods of MSA



- Exact methods of multiple alignment use **dynamic programming** and are guaranteed to find optimal solutions
- But they are not feasible for more than a few sequences
- The computational time required is $O(2^N L^M)$ where N is the number of sequences and L is the average sequence length
- The problem of MSA with minimum sum-of-pair (SP) score has been proven to be **NP-complete** (Lusheng Wang and Tao Jiang 1994)





Heuristic methods of MSA



- **Progressive methods:** Use a guide tree (related to a phylogenetic tree) to determine how to combine pairwise alignments one by one to create a multiple alignment
 - Example: ClustalW
- **Iterative approach:** Compute a sub-optimal solution and keep modifying that intelligently using dynamic programming or other methods until the solution converges
 - Example: IterAlign, Praline, MUSCLE
- **Consistency-based algorithms:** Use the information about the MSA as it is being generated to guide the pairwise alignments
 - For sequences x , y , and z , if residue x_i aligns with z_k and z_k aligns with y_j , then x_i should align with y_j
 - Example: T-COFFEE, Prrp, DiAlign, ProbCons, MAFFT
- **Structure-based methods:** Improve the accuracy of MSA by including information about the 3D structure of one or more members of a group of proteins being aligned
 - Example: Praline, T-COFFEE module Espresso





ClustalW





- It is a straightforward **progressive alignment strategy** where sequences are added one by one to the multiple alignment according to the order indicated by a pre-computed **guide tree**
- ClustalW has been implemented at many web sites (including EMBL/EBI), each with some different emphases:
 - <https://www.ebi.ac.uk/Tools/msa/clustalw2/>
 - <http://www.clustal.org/>
 - <https://www.genome.jp/tools-bin/clustalw>
 - https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_clustalw.html



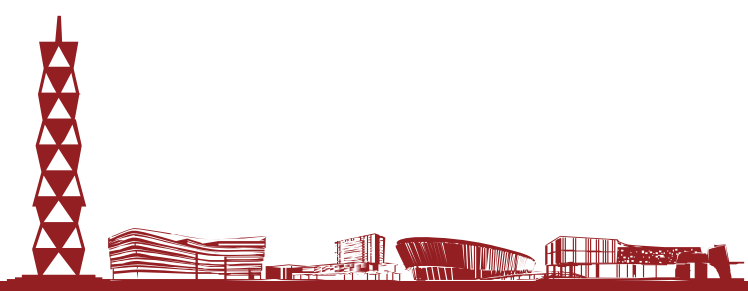


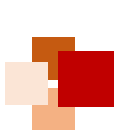
ClustalW: Outline of algorithm



- **Basic alignment procedure:**

1. Determine distances between sequences
2. Use a distance-based method to construct a guide tree (which is not exactly a phylogenetic tree) for the sequences
3. Add sequences to the growing alignment using the order given by the guide tree



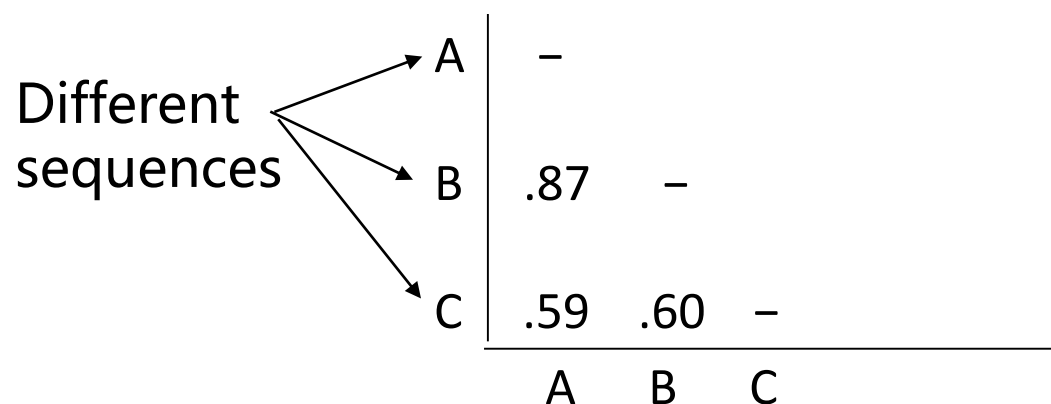


ClustalW: Step 1



• Step 1: Pairwise Alignments

Compare each sequence with each other and calculate a distance matrix.



Each number represents the number of exact matches divided by the sequence length (ignoring gaps). Thus, the higher the number, the more closely related the two sequences are.

In this distance matrix, sequence A is 87% identical to sequence B

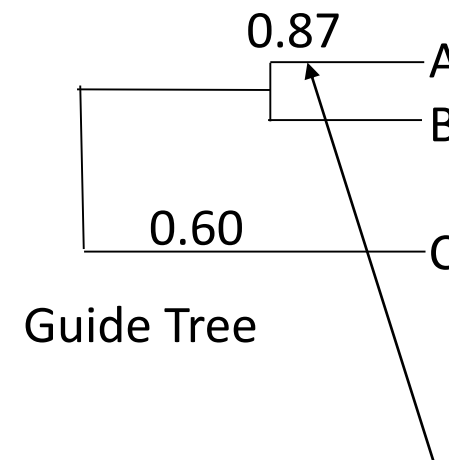
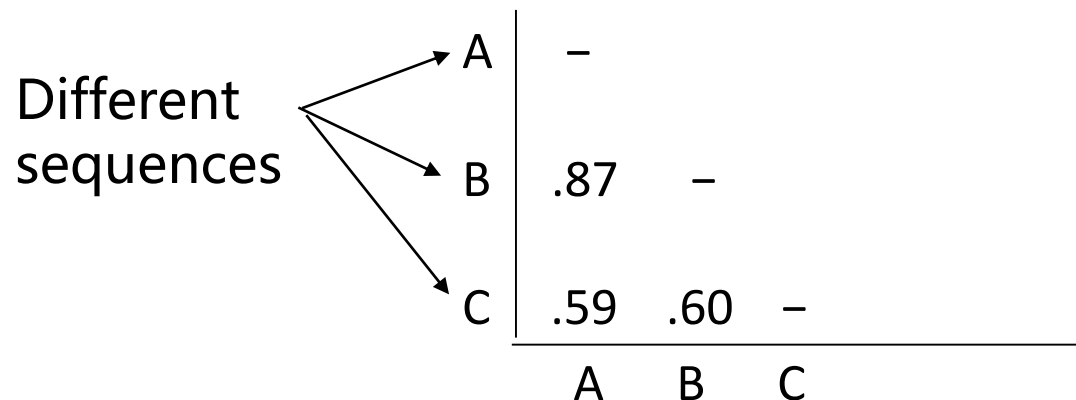


ClustalW: Step 2

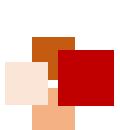


• Step 2: Create Guide Tree

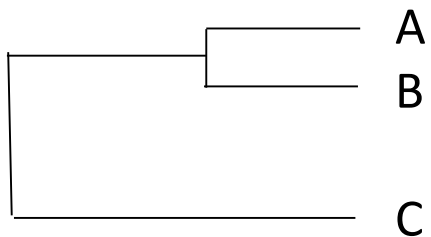
Use the Similarity (Distance) Matrix to create a Guide Tree to determine the “order” of the sequences to be added.



Branch length proportional to estimated divergence between A and B (0.13)



• Step 3: Progressive Alignment



Align A and B first. Then add sequence C to the previous alignment.

In the closely aligned sequences, gaps are given a heavier weight than more divergent sequences.

In ClustalW, as the alignment proceeds to longer branches the AA (amino acid) scoring matrices are changed to more divergent scoring matrices. The length of the branch is used to determine which matrix to use and contribute to the alignment score.



An example

Step 1: Building the Distance Matrix

$s(x,y) = +1$ if $x = y$; and -1 if $x \neq y$; $g = -2$

Using Dynamic Programming to produce the following induced pair wise alignments:

$M_{1,2}$

P	Y	R	F	T	I	K	S	M
P	Y	K	F	S	I	K	S	M

$score(M_{1,2}) = 5$

$M_{1,3}$

P	Y	R	F	T	I	K	S	M
P	Y	M	Y	S	S	E	S	M

$score(M_{1,3}) = -1$

$M_{1,4}$

P	Y	R	-	-	-	F	T	I	K	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$score(M_{1,4}) = -7$

S1:	P	Y	R	F	T	I	K	S	M			
S2:	P	Y	K	F	S	I	K	S	M			
S3:	P	Y	M	Y	S	S	E	S	M			
S4:	P	M	D	D	N	P	F	S	F	Q	S	M

similarity matrix

	M1	M2	M3	M4
M1				
M2	7/9= 0.78			
M3	4/9= 0.44			
M4	4/9= 0.44			

An example

Step 1: Building the Distance Matrix

$s(x,y) = +1$ if $x = y$; and -1 if $x \neq y$; $g = -2$

Using Dynamic Programming to produce the following induced pair wise alignments:

$M_{2,3}$

P	Y	K	F	S	I	K	S	M
P	Y	M	Y	S	S	E	S	M

$score(M_{2,3}) = 1$

$M_{2,4}$

P	Y	K	-	-	-	F	S	I	K	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$score(M_{2,4}) = -5$

$M_{3,4}$

P	Y	M	Y	-	-	-	S	S	E	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

$score(M_{3,4}) = -7$

S1:	P	Y	R	F	T	I	K	S	M			
S2:	P	Y	K	F	S	I	K	S	M			
S3:	P	Y	M	Y	S	S	E	S	M			
S4:	P	M	D	D	N	P	F	S	F	Q	S	M

similarity matrix

	M1	M2	M3	M4
M1				
M2	7/9= 0.78			
M3	4/9= 0.44	5/9= 0.56		
M4	4/9= 0.44	5/9= 0.56	4/9= 0.44	



An example



- Step 2: Create Guide Tree

Use the Similarity (Distance) Matrix to create a Guide Tree to determine the "order" of the sequences.

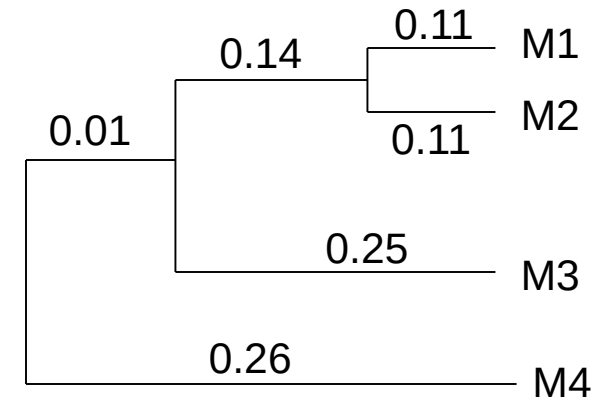
	M1	M2	M3	M4
M1				
M2	$7/9=0.78$			
M3	$4/9=0.44$	$5/9=0.56$		
M4	$4/9=0.44$	$5/9=0.56$	$4/9=0.44$	

Similarity Matrix

=>

	M1	M2	M3	M4
M1	0			
M2	0.22	0		
M3	0.56	0.44	0	
M4	0.56	0.44	0.56	0

Distance Matrix





An example



P	Y	R	F	T	I	K	S	M
P	Y	K	F	S	I	K	S	M

- Step 3:
- Multiple Sequence Alignment
- Dynamic Programming of M1,M2
- $s(x,y) = +1$ if $x = y$;
and -1 if $x \neq y$;
 $g = -2$

	0	P	Y	R	F	T	I	K	S	M
0	0	-2	-4	-6	-8	-10	-12	-14	-16	-18
P	-2	1	-1	-3	-5	-7	-9	-11	-13	-15
Y	-4	-1	2	0	-2	-4	-6	-8	-10	-12
K	-6	-3	0	1	-1	-3	-5	-5	-7	-9
F	-8	-5	-2	-1	2	0	-2	-4	-6	-8
S	-10	-7	-4	-3	0	1	-1	-3	-3	-5
I	-12	-9	-6	-5	-2	-1	2	0	-2	-4
K	-14	-11	-8	-7	-4	-3	0	3	1	-1
S	-16	-13	-10	-9	-6	-5	-2	1	4	2
M	-18	-15	-12	-11	-8	-7	-4	-1	2	5





An example

- Step 3:
- Multiple Sequence Alignment
- Dynamic Programming of (M1,M2),M3
- $s(x,y) = +1$ if $x = y$;
and -1 if $x \neq y$; $g = -2$

P	Y	R	F	T	I	K	S	M
P	Y	K	F	S	I	K	S	M
P	Y	M	Y	S	S	E	S	M

	0	P	Y	R	F	T	I	K	S	M
0	0	-3	-6	-11	-14	-19	-22	-25	-28	-31
P	-4	3	0							
Y	-8	-1	6							
M	-12									
Y	-16									
S	-20									
S	-24									
E	-28									
S	-32									
M	-36									

M1: P M1,M2= 1
M2: P M1,M3= -2
M3: - M2,M3= -2
Score: 1 - 2 - 2 = -3

M1: PY M1,M2= 2
M2: PY M1,M3= 2
M3: PY M2,M3= 2
Score: 2 + 2 + 2 = 6





An example

- Step 3:
- Multiple Sequence Alignment
- Dynamic Programming of (M1,M2),M3
- $s(x,y) = +1$ if $x = y$;
and -1 if $x \neq y$; $g = -2$

P	Y	R	F	T	I	K	S	M
P	Y	K	F	S	I	K	S	M
P	Y	M	Y	S	S	E	S	M

	0	P	Y	R	F	T	I	K	S	M
0	0	-3	-6	-11	-14	-19	-22	-25	-28	-31
P	-4	3	0	-5	-8	-13	-16	-19	-22	-25
Y	-8	-1	6	1	-2	-7	-10	-13	-16	-19
M	-12	-5	2	3	0	-5	-8	-11	-14	-17
Y	-16	-9	-2	-1	2	-3	-6	-9	-12	-15
S	-20	-13	-6	-5	-2	1	-2	-5	-8	-11
S	-24	-17	-10	-9	-6	-3	0	-3	-6	-9
E	-28	-21	-14	-13	-10	-7	-4	-1	-4	-7
S	-32	-25	-18	-17	-14	-11	-8	-5	2	-1
M	-36	-29	-22	-21	-18	-15	-12	-9	-2	5



An example

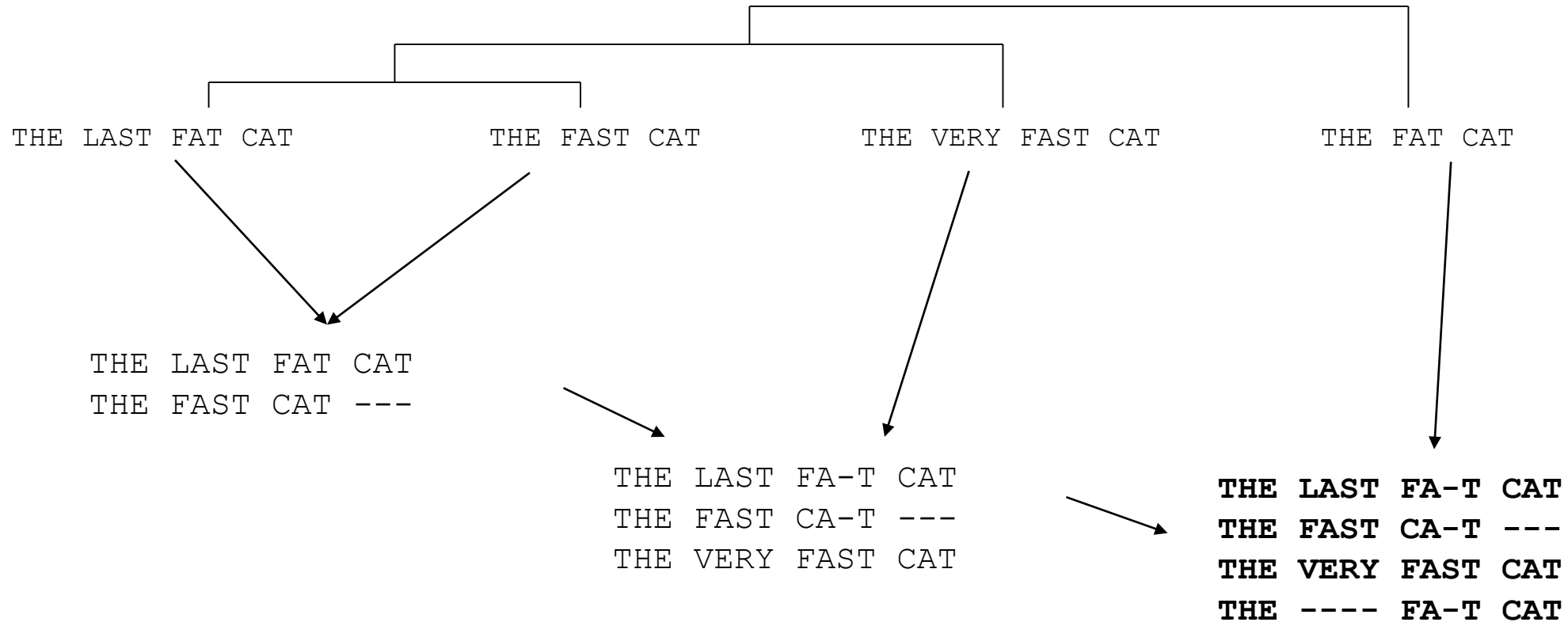
- Step 3:
- Multiple Sequence Alignment
- Dynamic Programming of (M1,M2,M3),M4
- $s(x,y) = +1$ if $x = y$;
and -1 if $x \neq y$; $g = -2$

P	Y	R	-	-	-	F	T	I	K	S	M
P	Y	K	-	-	-	F	S	I	K	S	M
P	Y	M	-	-	-	Y	S	S	E	S	M
P	M	D	D	N	P	F	S	F	Q	S	M

	0	P	Y	R	F	T	I	K	S	M
		P	Y	K	F	S	I	K	S	M
		P	Y	M	Y	S	S	E	S	M
0	0	-3	-6	-15	-22	-29	-36	-43	-46	-49
P	-6	6	3	-6	-13	-20	-27	-34	-37	-40
M	-12	0	6	-3	-10	-17	-24	-31	-34	-37
D	-18	-6	0	0	-7	-14	-21	-28	-31	-34
D	-24	-12	-6	-6	-4	-11	-18	-25	-28	-31
N	-30	-18	-12	-12	-10	-8	-15	-22	-25	-28
P	-36	-24	-18	-18	-16	-14	-12	-19	-22	-25
F	-42	-30	-24	-24	-18	-20	-18	-16	-19	-22
S	-48	-36	-30	-30	-24	-18	-24	-22	-10	-13
F	-54	-42	-36	-36	-30	-24	-22	-28	-16	-10
Q	-60	-48	-42	-42	-36	-30	-28	-26	-22	-16
S	-66	-54	-48	-48	-42	-36	-34	-32	-20	-22
M	-72	-60	-54	-54	-48	-42	-40	-38	-26	-14



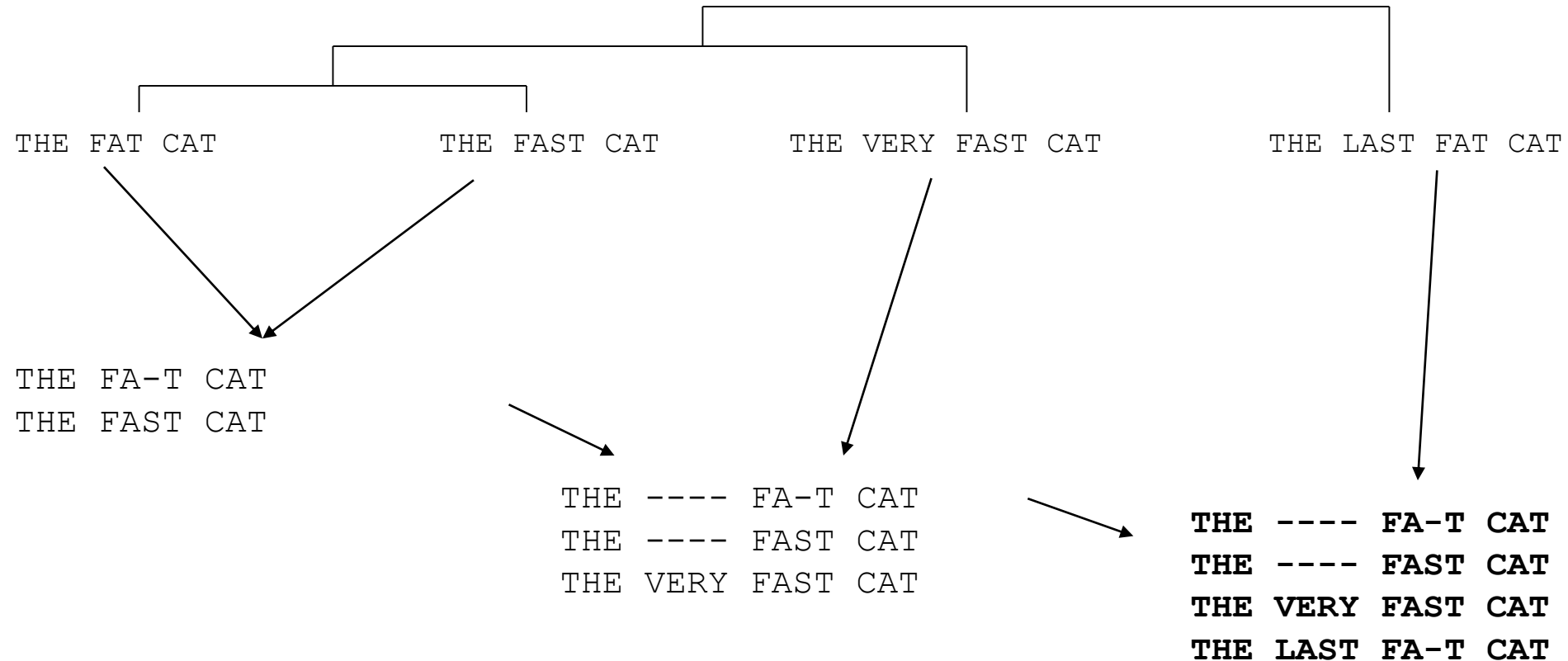
Progressive MSA stage 3 of 3: progressively align the sequences following the branch order of the tree: Order matters



Adapted from C. Notredame, Pharmacogenomics 2002



Progressive MSA stage 3 of 3: progressively align the sequences following the branch order of the tree: Order matters



Adapted from C. Notredame, Pharmacogenomics 2002



Additional features of ClustalW improve its ability to generate accurate MSAs

- Individual weights are assigned to sequences; very **closely** related sequences are given **less weight**, while **distantly** related sequences are given **more weight**
- Scoring matrices are varied dependent on the presence of conserved or divergent sequences, e.g.:

PAM20	80-100% id
PAM60	60-80% id
PAM120	40-60% id
PAM350	0-40% id

- Residue-specific gap penalties are applied





“Once a gap, always a gap” rule

- There are many possible ways to make an MSA
- Where gaps are added is a critical question
 - Gaps are often added to the first two (closest) sequences
 - To change the initial gap choices later on means to give more weight to distantly related sequences
 - To maintain the initial gap choices is to trust that those gaps are most believable

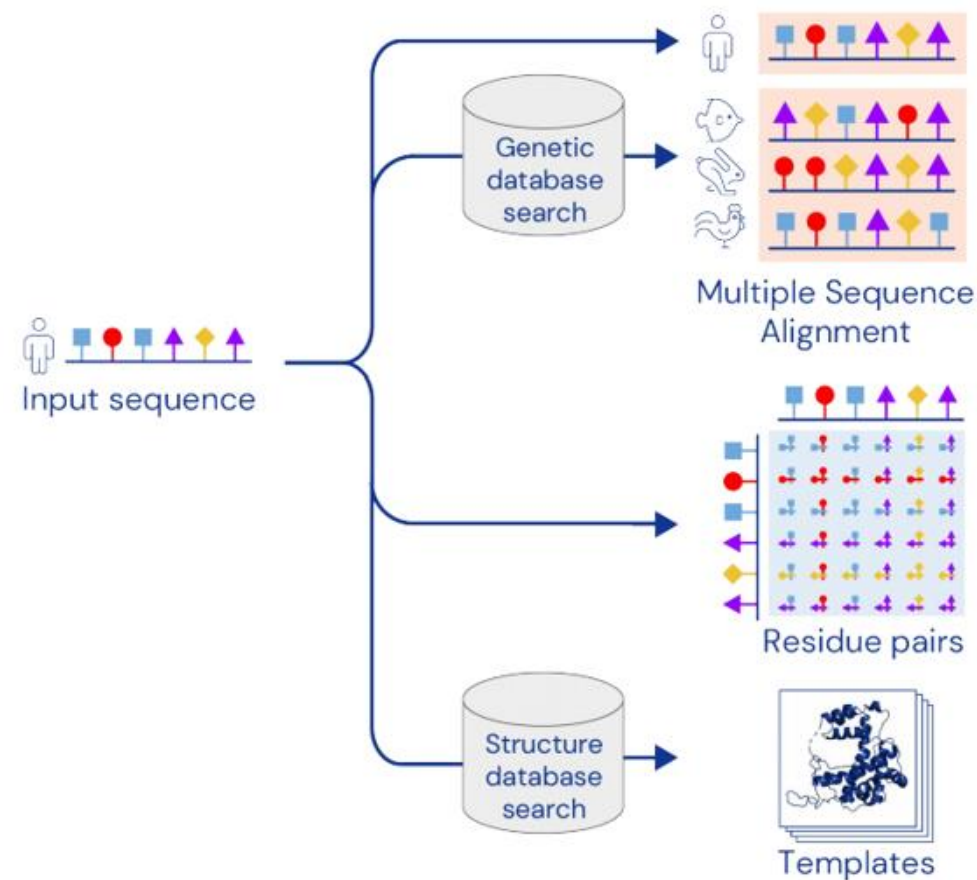
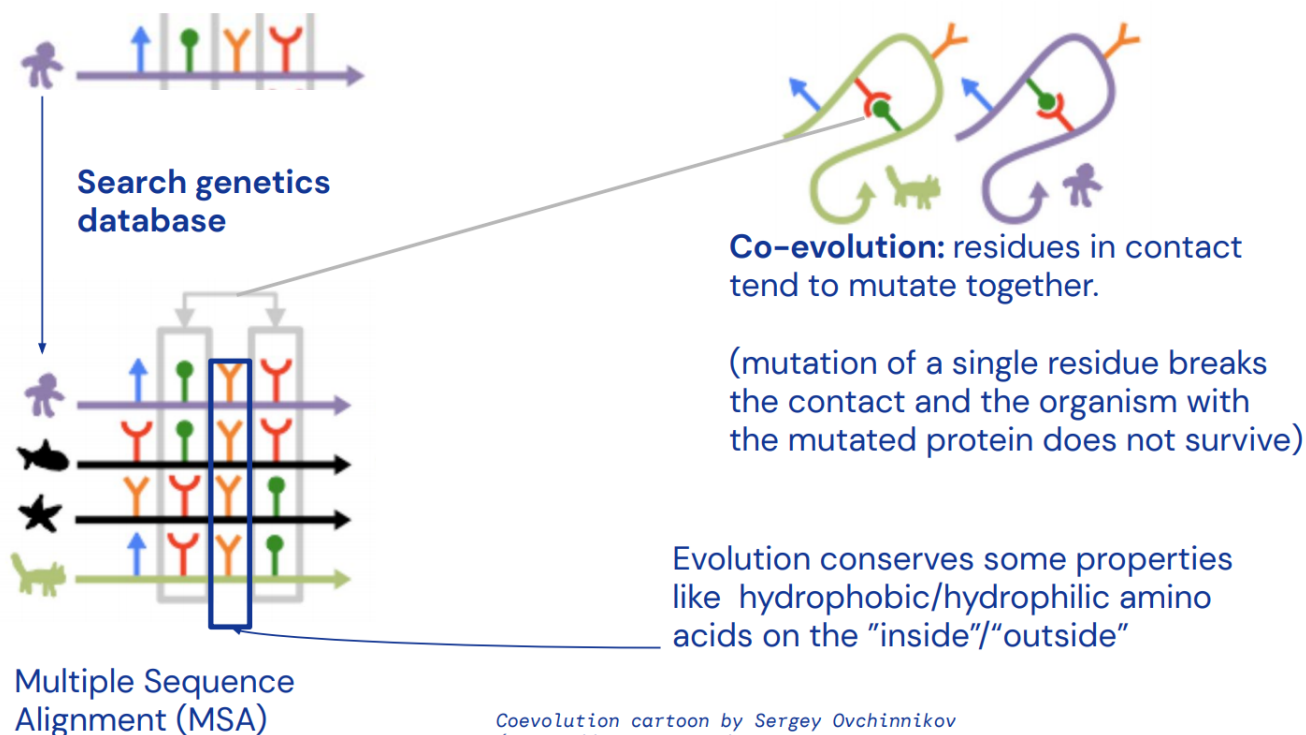




MSA Transformer



Recap: importance of MSA for AlphaFold



A key intuition of AlphaFold2: To determine protein structure from evolution

Three types of input to AlphaFold2

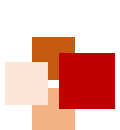


Two paradigms of unsupervised learning from protein sequences

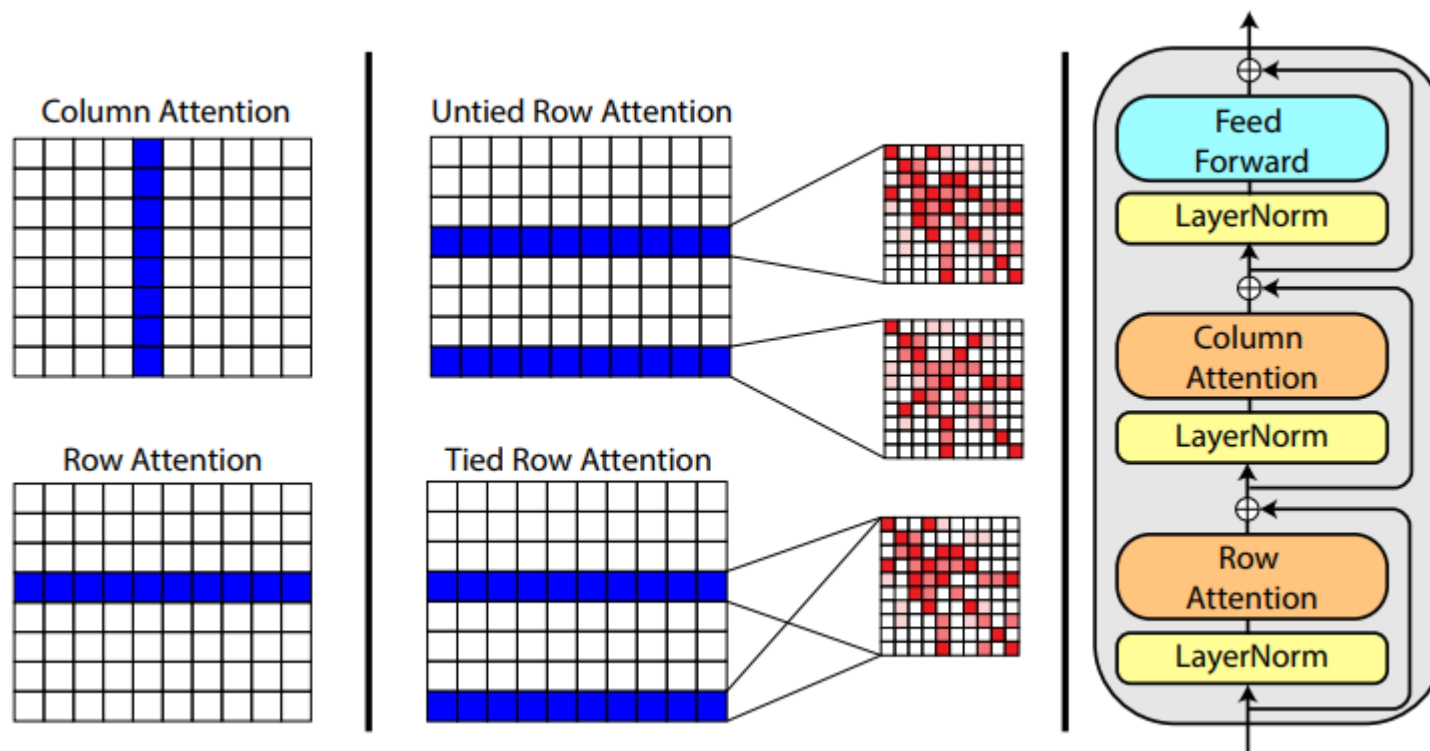


- **Paradigm 1:** To make inferences from a family of evolutionary related sequences (i.e. an MSA) by fitting a model to **each family** independently
 - E.g. to fit a Markov Random Field in the form of a **Potts model** to a family of aligned sequences to extract co-evolutionary signals (Lapedes et al. 1999)
- **Paradigm 2:** Unsupervised protein language models (PLMs) trained across millions of diverse sequences to learn structures and functions of proteins
 - To fit large neural networks with shared parameters **across millions of diverse sequences**, rather than fitting a model separately to each family of sequences
 - E.g. The series of Transformer-based PLMs named **ESM (Evolutionary Scale Modeling)**, by Meta (Facebook),
<https://github.com/facebookresearch/esm>



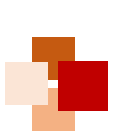


Overview of MSA Transformer



- MSA Transformer combines the aforementioned two paradigms of unsupervised learning of protein structures from sequences
- Interleaving row and column attention across the input MSAs (i.e. using **axial attention**)
- **Tied row attention**: sharing a single attention map across the rows
- Trained with a variant of the masked language modeling objective across many protein families





Results on unsupervised contact prediction

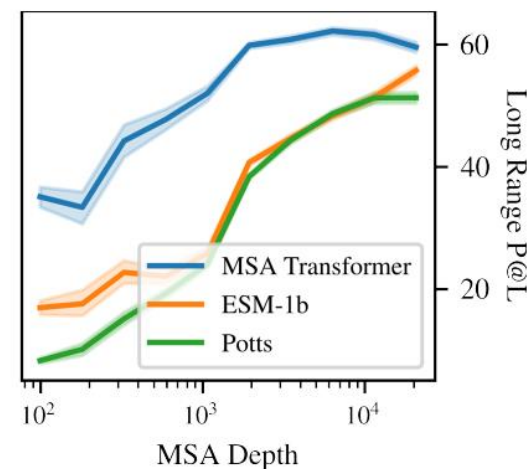
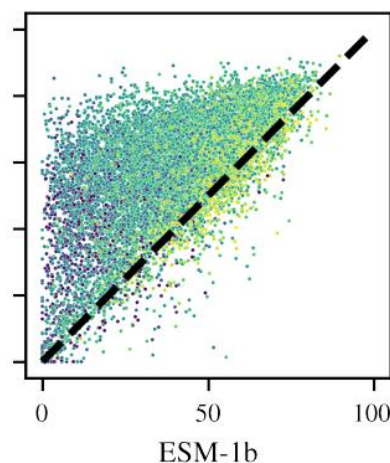
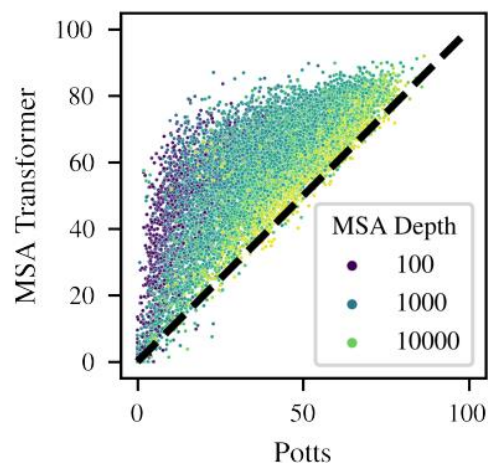


Table 1. Average long-range precision for MSA and single-sequence models on the unsupervised contact prediction task.

Model	L	L/2	L/5
Potts	39.3	52.2	62.8
TAPE	11.2	14.0	17.9
ProTrans-T5	35.6	46.1	57.8
ESM-1b	41.1	53.3	66.1
MSA Transformer	57.7	72.5	83.1

Table 2. Unsupervised contact prediction on CASP13 and CAMEO (long-range precision). Note the large improvement of MSA Transformer over classical Potts models and ESM-1b.

Model	CASP13-FM		CAMEO	
	L	L/5	L	L/5
ProTrans-T5	16.6	26.7	25.3	42.6
ESM-1b	16.8	30.1	30.7	52.3
Potts	17.0	31.8	23.9	42.7
MSA Transformer	42.7	65.3	40.5	59.1



Top-L long-range contact precision (higher is better)





References



- R. Durbin, S. Eddy, A. Krogh, G. Mitchison (1998). "Biological sequence analysis: Probabilistic models of proteins and nucleic acids" , Chapter 5 and Chapter 6, Cambridge University Press.
- Jonathan Pevsner (2015). "Bioinformatics and Functional Genomics " (3rd Edition), Chapter 6 "Multiple Sequence Alignment" .
- Wing-Kin Sung (2010). "Algorithms in Bioinformatics: A Practical Introduction" , Chapter 6 "Multiple Sequence Alignment" .
- D. G. Higgins and P. M. Sharp (1988). "CLUSTAL: A package for performing multiple sequence alignment on a microcomputer" , Gene, 73(1):237–244.
- J. D. Thompson, et al. (1994). "CLUSTAL W: Improving the sensitivity of progressive multiple sequence alignment through sequence weighting, positions-specific gap penalties and weight matrix choice" , Nucleic Acids Research, 22:4673–4680.
- Remmert M, Biegert A, Hauser A, Söding J (2012). "HHblits: Lightning-fast iterative protein sequence searching by HMM-HMM alignment.". Nature Methods, 9 (2): 173–175.
- Rao R., et al. (2021). "MSA Transformer" , Proceedings of the 38th International Conference on Machine Learning (ICML), **139**, 8844-8856.





Recap



- In this lecture, we have learned:
 - The definition, properties and features of multiple sequence alignment
 - Two main scoring functions for evaluating MSA (i.e. sum-of-pairs, distance to consensus)
 - Exact method and a few heuristic approaches to MSA: their ideas and example software tools
 - The ideas and main steps of the ClustalW
 - MSA Transformer is a unsupervised protein language model for various protein structure prediction tasks





Appendix: HHblits





Profile HMMs



- Models for (amino acid) sequence families
- Special structure:
 - match, insert, delete states
 - Special transition structure
- Parameters are estimated from a given multiple sequence alignment
- Can be used for examining the relation of new sequences to the family represented by the **profile**
- Special cases of profiles:
 - PWMs (position weight matrices)
 - PSSMs (position specific scoring matrices)

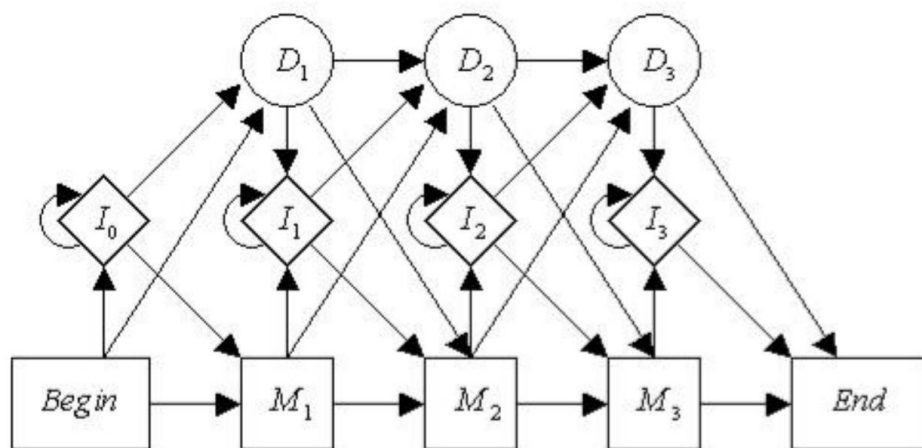




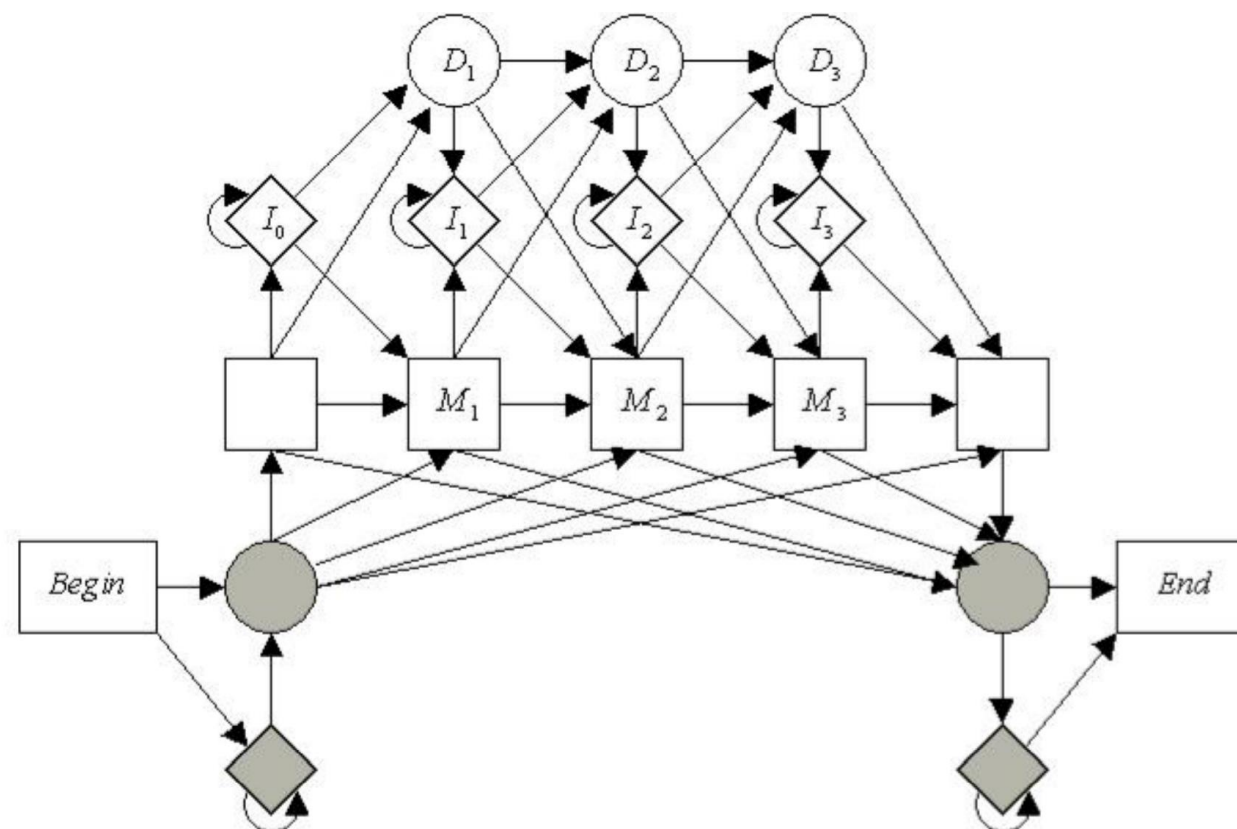
Profile HMM: standard structure



- All HMM algorithms (Viterbi, Forward, Backward, Baum-Welch, etc.) can be adapted for the profile HMM

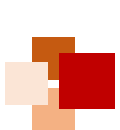


Profile HMM for global alignment



Profile HMM for local alignment





Learning profile HMMs from MSAs



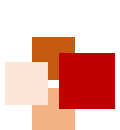
- **Input:** Multiple alignment X of some sample sequences from a sequence family to be modeled by the profile HMM
- **Step 1:** Select some columns $1, \dots, L$ of the alignment X to the backbone; these columns will correspond to the match states $M_1, \dots, M_L$ of the profile HMM
 - Take the best conserved columns, with no gap
- **Step 2:** Estimate probabilities $a_{kl}, e_k(b)$

$$a_{kl} = \frac{A_{kl}}{\sum_{q \in Q} A_{kq}} \quad e_k(b) = \frac{E_k(b)}{\sum_{\sigma \in \Sigma} E_k(\sigma)}$$

where

- A_{kl} = (the count of transitions $k \rightarrow l$ in X) + 1 (Laplace rule of pseudocounts)
- $E_k(b)$ = (the count of emissions of b from state k in X) + 1

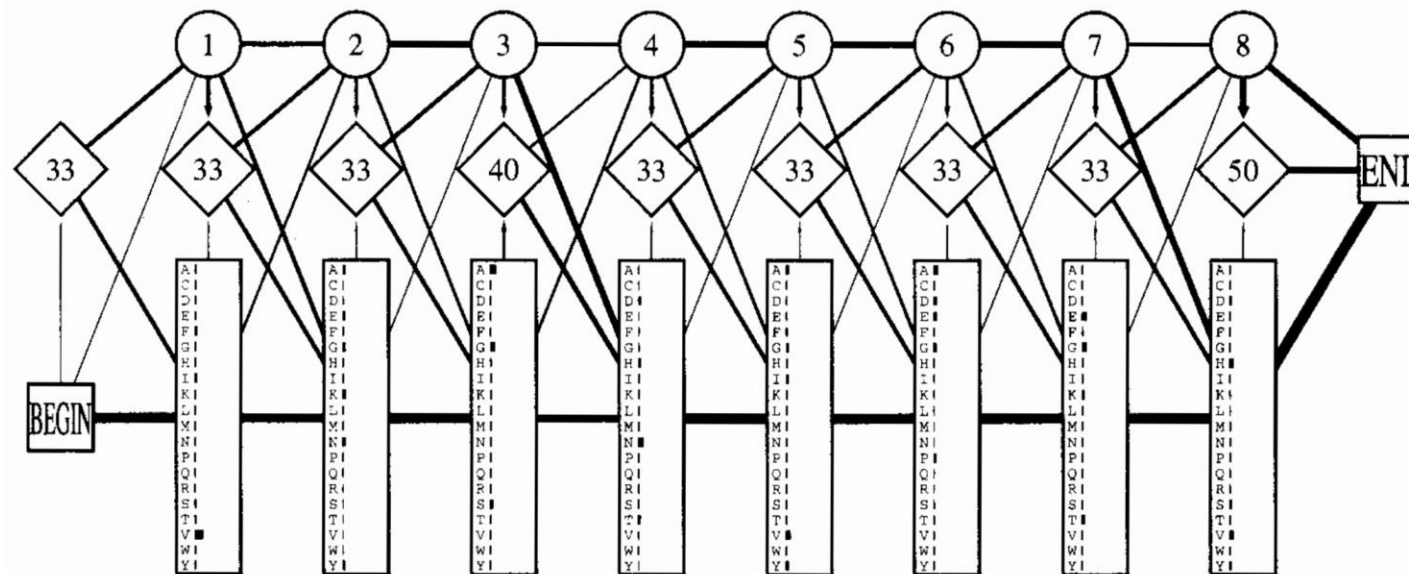




Learning a profile HMM: an example



```
HBA_HUMAN    ...VGA--HAGEY...
HBB_HUMAN    ...V----NVDEV...
MYG_PHYCA    ...VEA--DVAGH...
GLB3_CHITP    ...VKG-----D...
GLB5_PETMA    ...VYS--TYETS...
LGB2_LUPLU    ...FNA--NIPKH...
GLB1_GLYDI    ...IAGADNGAGV...
          ***  *****
```



- 10 columns from the MSA of 7 globin protein sequences
- The starred columns are ones that will be treated as “matches” in the profile HMM

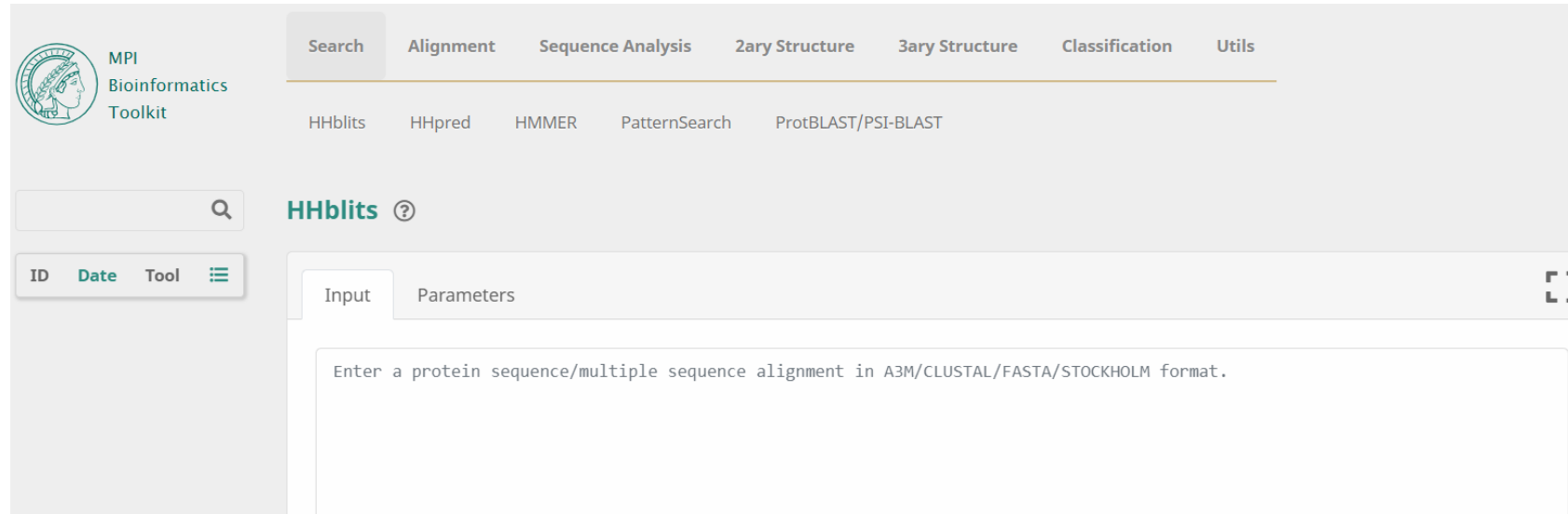
- An HMM derived from the MSA using **Laplace's rule** (i.e. add pseudocount 1 to each count)
- **Emission probabilities** are shown as bars opposite the different amino acids for each match state
- **Transition probabilities** are indicated by the thickness of the lines
- The **probabilities of $I \rightarrow I$ transitions for inserts** are shown as percentages in the insert states





HHblits: lightning-fast iterative protein sequence searching by HMM-HMM alignment

Web Server: <https://toolkit.tuebingen.mpg.de/tools/hhblits>



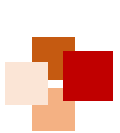
GitHub: <https://github.com/soedinglab/hh-suite>

HH-suite3 for sensitive sequence searching

(C) Johannes Soeding, Markus Meier, Martin Steinegger, Milot Mirdita, Michael Remmert, Andreas Hauser, Andreas Biegert

BioConda install 7.2k BioContainer 90k+ downloads 4.3k docker pulls 609 Azure Pipelines succeeded

The HH-suite is an open-source software package for sensitive protein sequence searching based on the pairwise alignment of hidden Markov models (HMMs).

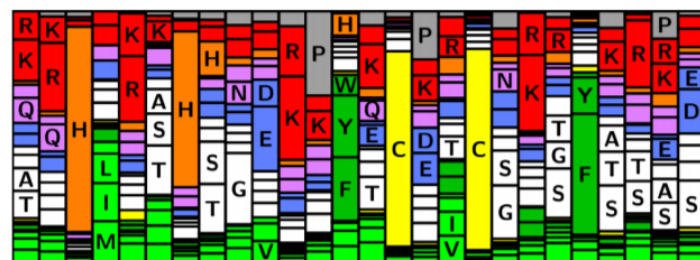


HHblits translates an alignment into a sequence of column states

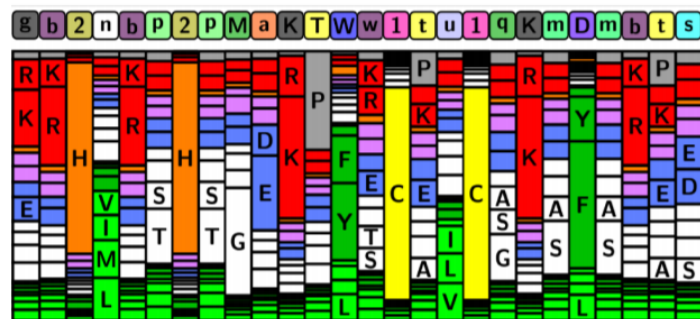
TRHIRIHTGQKPFQCRICMRNFSRSD
KRHIRIHTGEKPFQCRICGRNFSRKD
KKHIRTHTGDRPYKCTDCPKTFICAG
CSHQKTHHTTERPYICRNCGKTFRFC
SMHERVHDASRIYQCEYCPKSYVTPS

MSA

⋮



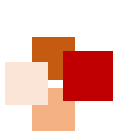
HMM profile



An alphabet of 219 letters

Nature Methods **9**, 173–175 (2012)

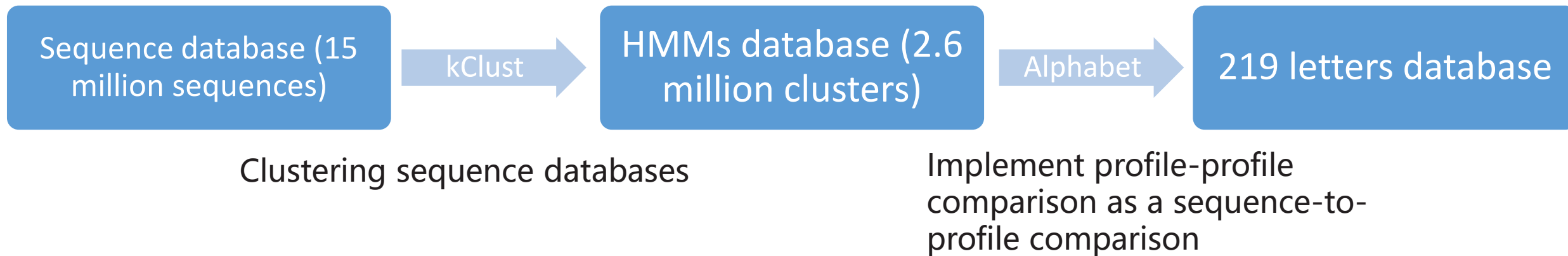


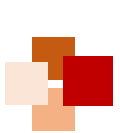


HHblits needs a database of HMMs that covers the entire sequence space

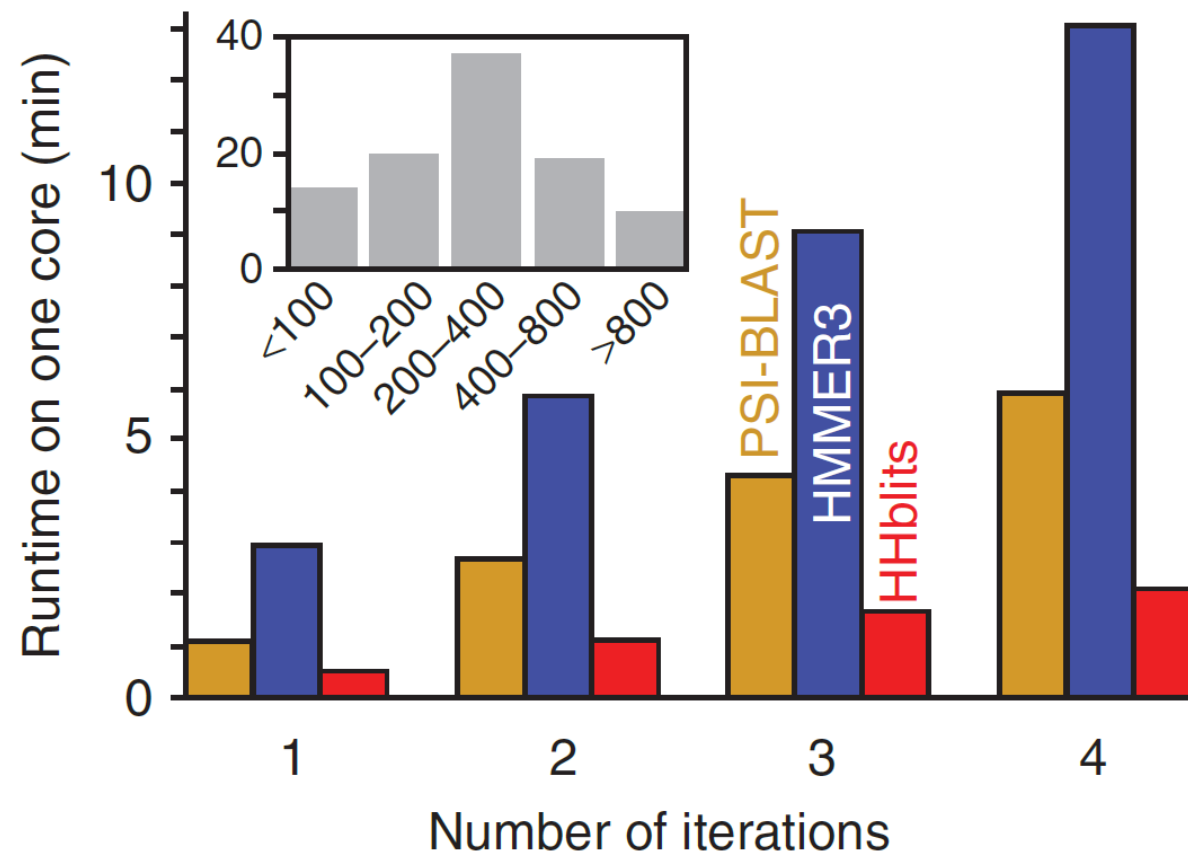
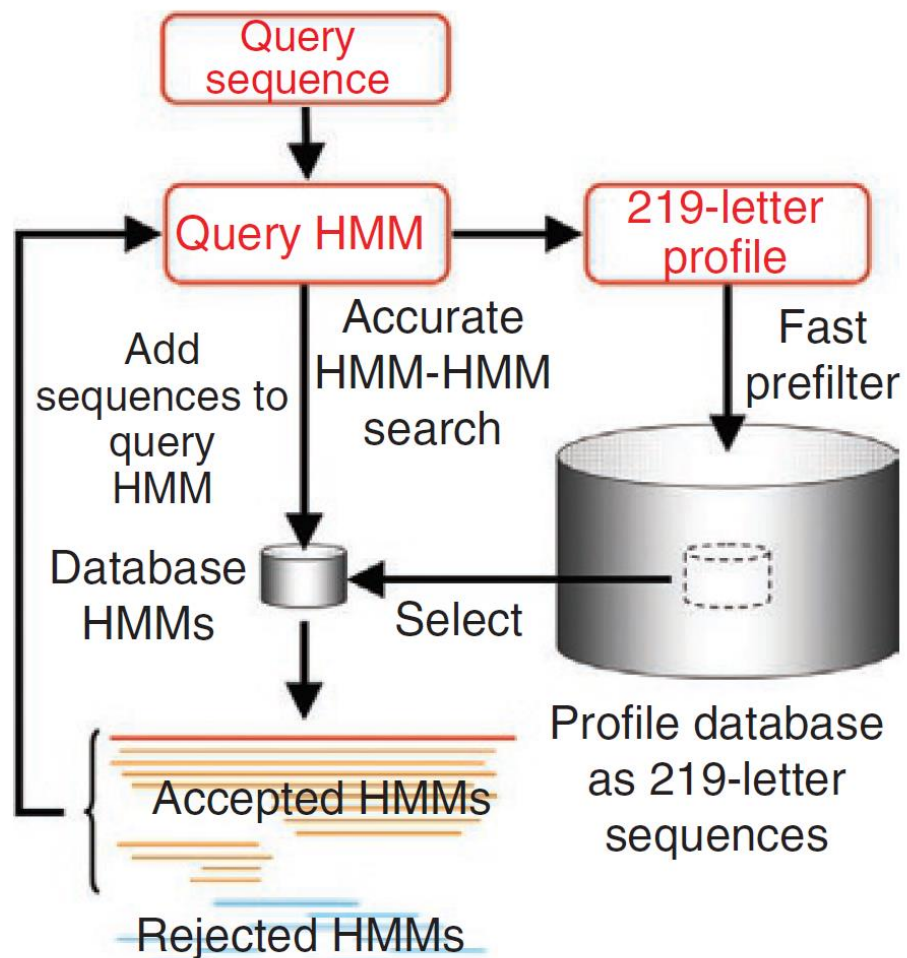


e.g. UniPort20_2012_06



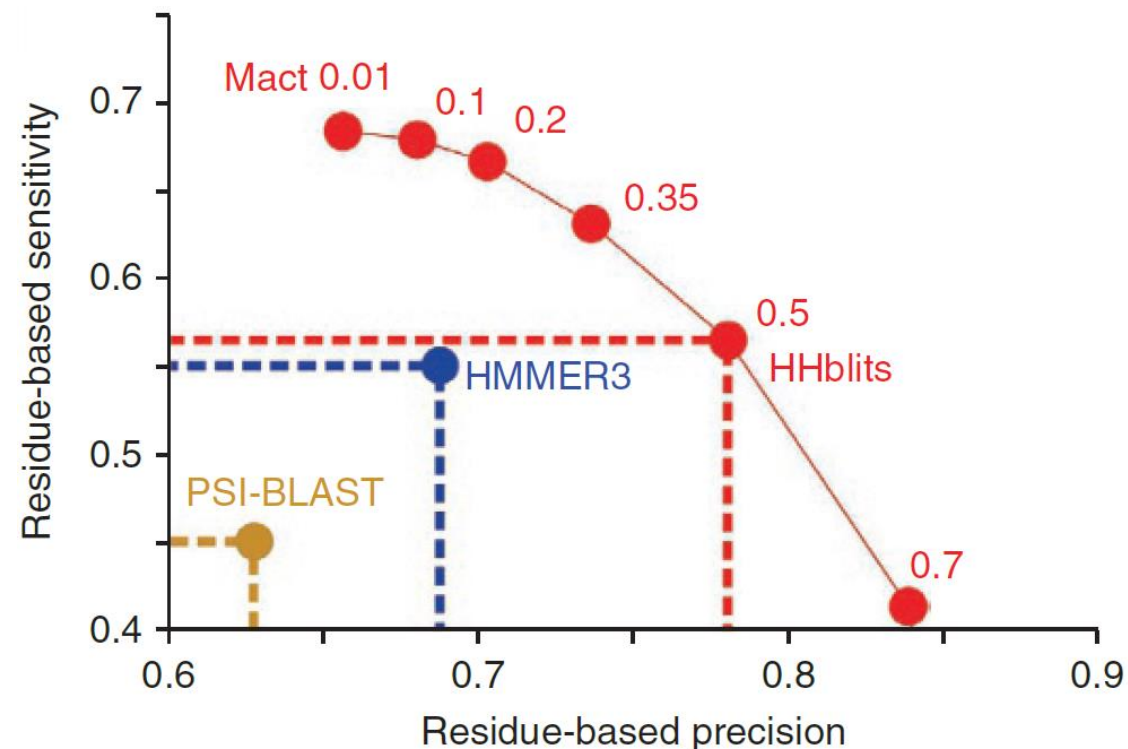
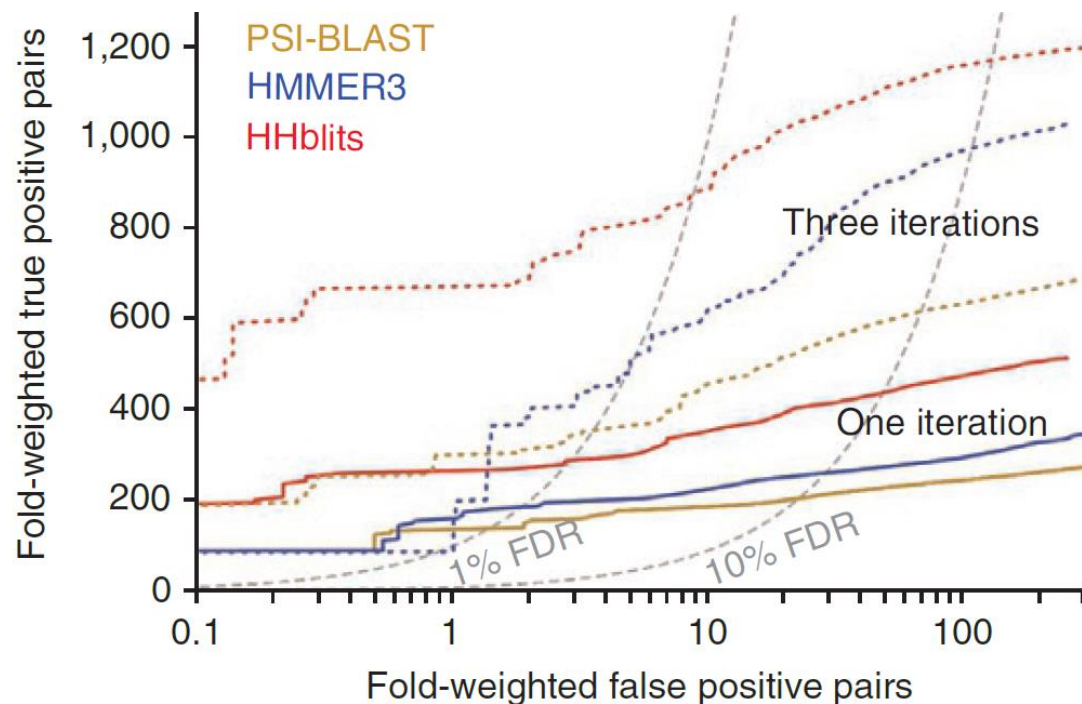


HHblits workflow and speed comparison



One iteration	HHblits	PSI-BLAST	HMMER3
Median	31 s	1 min 7 s	2 min 57 s
Average	1 min 13 s	1 min 26 s	5 min 8 s

Sensitivity in detecting homologous proteins and quality of the pairwise alignments



Nat Methods 9, 173–175 (2012)